UPI Fraud Stress Lab

Campaign-level evaluation before intervention

An auditable engineering case study, not a deployed UPI detector.

Working software, matched synthetic controls and preserved adverse results.

Academic review draft • author, venue and external checker approval pending

Project records: results/ and docs/VERIFICATION_REPORT.md

Why evaluate a campaign?

Transaction score

AUC ranks examples in a specified dataset.

It does not say when a sequence stops.

Campaign endpoint

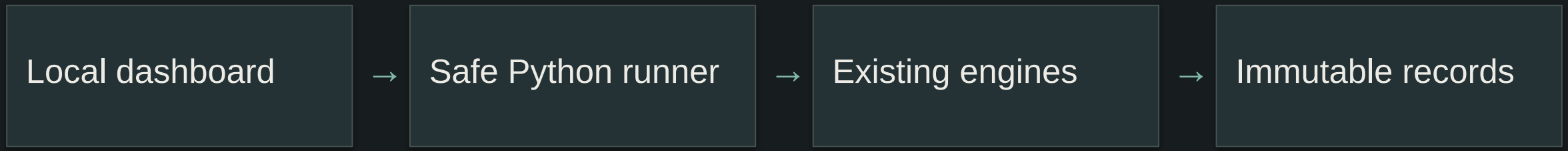
Does a simulated actor reach its target before BLOCK?

FLAG can still transfer; BLOCK terminates.

Gross simulated debit is not victim loss or proven savings.

Project records: results/ and docs/VERIFICATION_REPORT.md

The implemented evidence pipeline



Independent checkers and raw-record reconstruction feed tables, figures and the paper.

No arbitrary shell, public binding or Git/Gate C endpoint.

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A narrow, registered exploratory pilot

Data and detectors

9 independent blocks; 18 fresh detector fits.

7,200 evaluated attack campaigns.

HVB allowed; declared GE excluded from development.

Ordinary benign and GE-mimic test strata.

Five online-matched controls

Static; rule-adaptive; trained RL;

explicit-label-blind RL; zero Q.

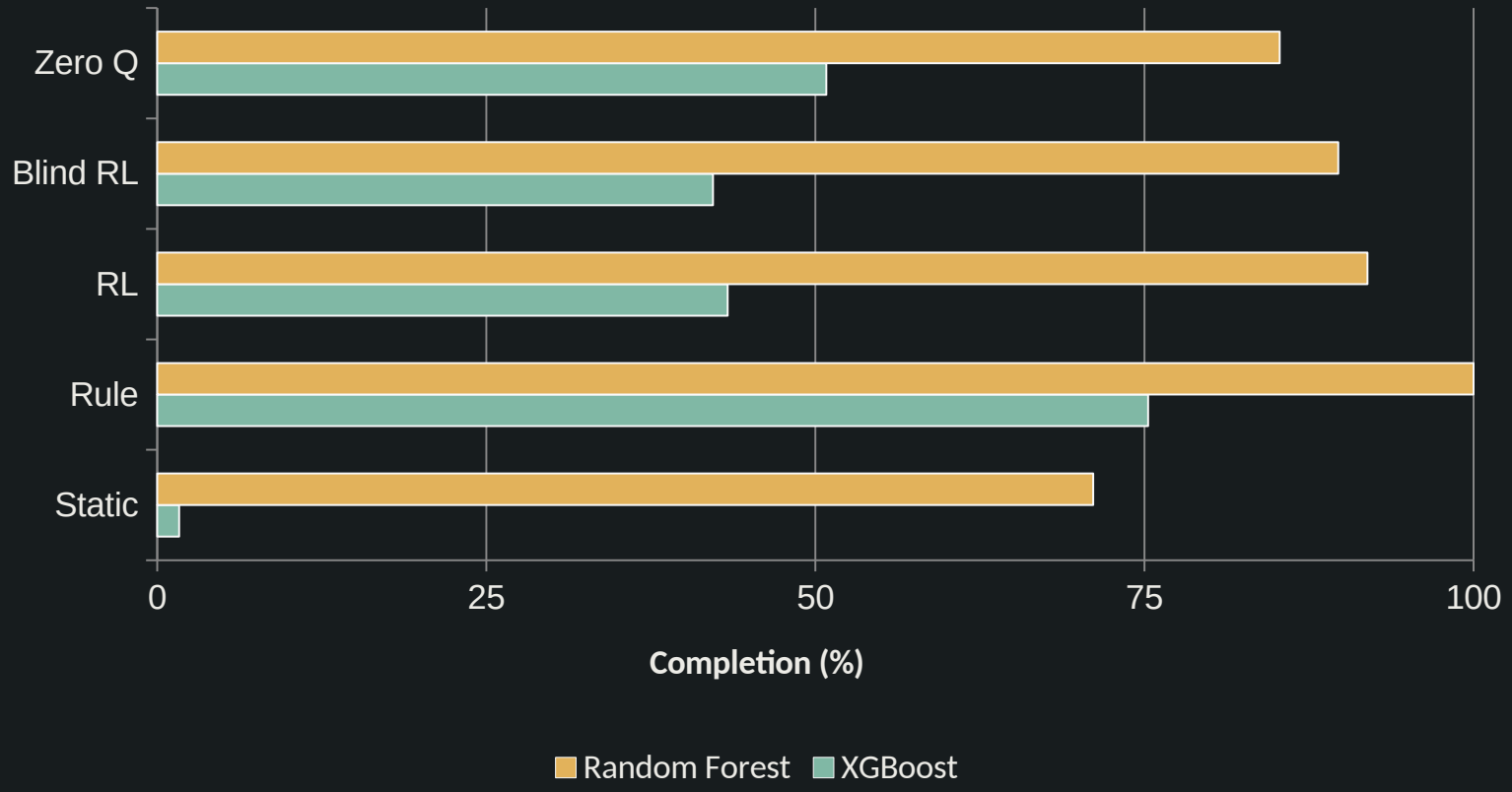
Common actions, clipping, pacing and budgets.

Offline training costs remain unequal.

No powered confirmatory lockbox and no real-world UPI validation.

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Completion depends on detector and control



Held-out GE

360 campaigns per arm/detector

9 independent blocks

Block SD (percentage points)

Arm: XGB / RF

Static: 1.25 / 7.61

Rule: 14.33 / 0.00

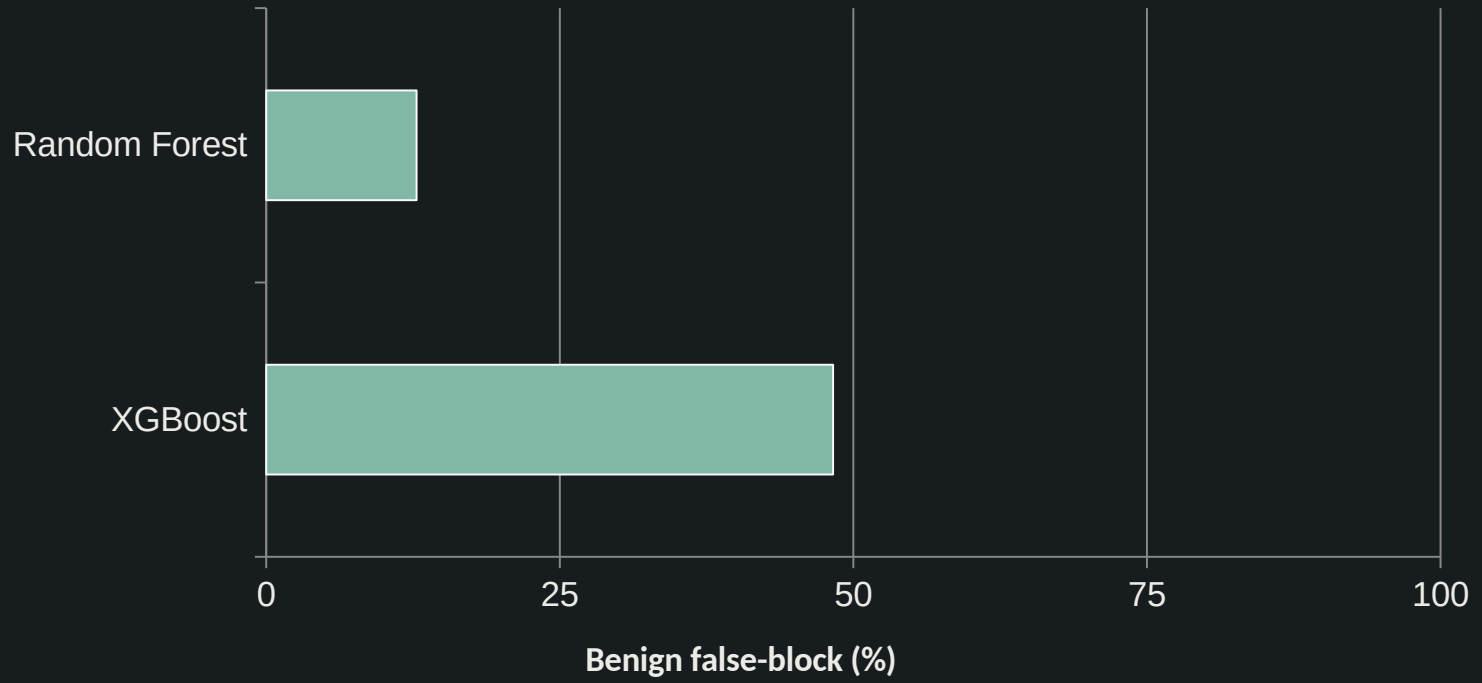
RL: 24.08 / 8.36

Blind RL: 24.86 / 9.56

Zero Q: 17.98 / 6.43

Project records: results/ and docs/VERIFICATION_REPORT.md

Benign mechanism shift is a major failure



48.278%

1,738 / 3,600 XGBoost episodes

12.806%

461 / 3,600 RF episodes

Ordinary benign: 0 observed BLOCK / 3,600 per detector.

No population guarantee; nine blocks. Mechanism exclusion ≠ reliable discrimination.

Project records: results/ and docs/VERIFICATION_REPORT.md

What we can defensibly contribute

Prior research already does

Sequence fraud detection, synthetic finance,

banking evasion and learned attackers.

This project is not the first RL fraud attack.

Implemented contribution

Auditable campaign/ledger infrastructure,

five matched controls, narrow lineage checks,

benign-shift controls and reconstruction.

No universal superiority claim.

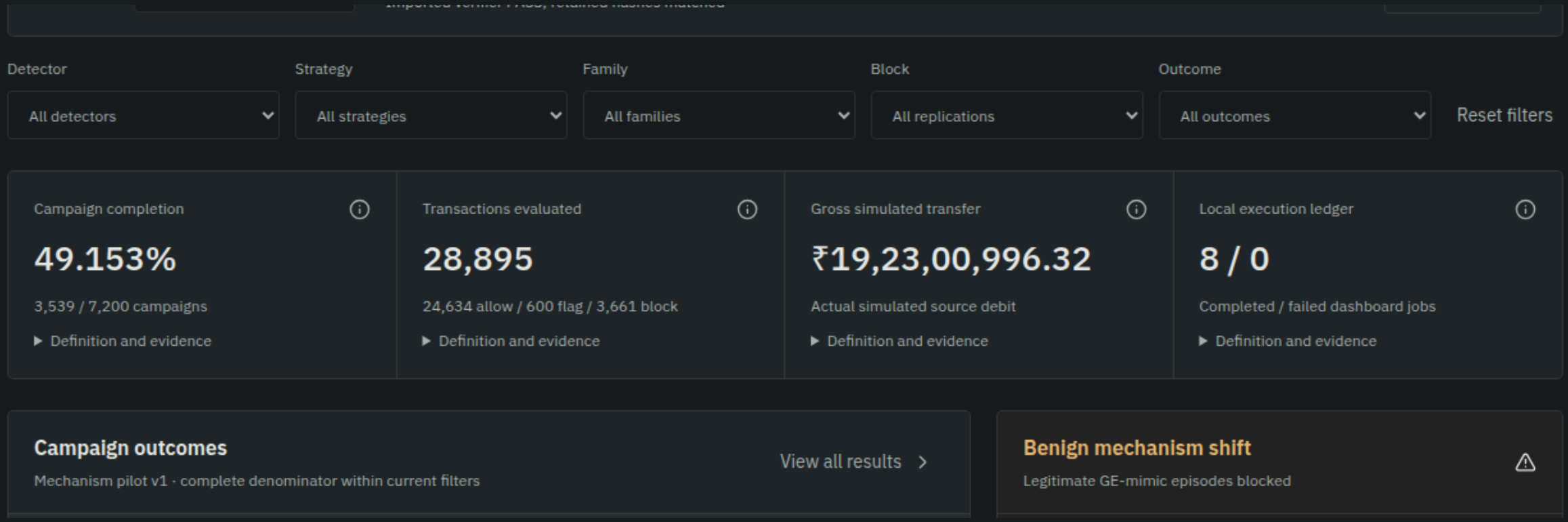
Fraud-RLA

Banking evasion

PaySim

Source: verified literature links below; project implementation and retained results

A real local research workstation



Actual overview detail. Live demo: run → logs → campaign → comparisons → export.

Project records: results/ and docs/VERIFICATION_REPORT.md

Verified engineering; bounded scientific claims

Executed verification

478 distinct tests passed in fresh Python 3.11.

Short legacy/pilot main, repeat, reload and check.

1,012,673 retained-pilot verifier checks;

30,985 TD updates replayed.

Still requires human action

Approve authors, venue and AI-use policy.

Obtain genuine approved AI/similarity reports.

Select a project license.

Do not infer real-world effectiveness.

Ready for exploratory academic review, not unqualified submission.

Scientific validity extends only as far as the evidence. Gate C remains HOLD.

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